Naive Bayes classifier

$P(\text{Play-Tennis} = \text{Yes}) = 9/14 = 0.64$

$P(\text{PlayTennis} = \text{no}) = 5/14 = 0.36$

Outlook

Sunny $\Rightarrow$ Yes $= 2/9$
$\Rightarrow$ No $= 3/5$

overcast $\Rightarrow$ Yes $= 4/9$
$\Rightarrow$ No $= 0$

rain $\Rightarrow$ Yes $= 3/9$
No $= 2/5$

Temperature

hot $\Rightarrow$ Yes $= 2/9$
$\Rightarrow$ No $= 2/5$

mild $\Rightarrow$ Yes $= 4/9$
$\Rightarrow$ No $= 2/5$

cool $\Rightarrow$ Yes $= 3/9$
$\Rightarrow$ No $= 1/5$

Humidity

high $\Rightarrow$ Yes $= 3/9$
~~normal~~ high $\Rightarrow$ No $= 4/5$

normal $\Rightarrow$ Yes $= 6/9$
$\Rightarrow$ No $= 1/5$

windy

~~win~~ Strong $\Rightarrow$ yes $= 3/9$
$\Rightarrow$ No $= 3/5$

weak $\Rightarrow$ yes $= 6/9$
No $= 2/5$

Question

(Outlook = Sunny, Temperature = cool, Humidity = high

Wind = Strong)

$V_{NB} = \arg\max P(V_j) \prod_i P(a_i / V_j)$

$= \arg\max (P(V_i))\, P(\text{outlook} = \text{Sunny} / V_j)$

$P(\text{Humidity} = \text{high} / V_j)$

~~P (outl~~

$P(\cancel{\text{cool}}\ \text{Temperature} = \text{cool} / V_j)$

$P(\text{wind} = \text{Strong} / V_i)$

$V_{NB}(\text{yes}) = P(\text{yes})\, P(\text{Sunny} / \text{yes})\, P(\text{cool} / \text{yes})$

$P(\text{high} / \text{yes})\, P(\text{Strong} / \text{yes}) = 0.0053$

$V_{NB}(\text{No}) = P(\text{No})\, P(\text{Sunny} / \text{No})\, P(\text{cool} / \text{No})$

$P(\text{high} / \text{No})\, P(\text{Strong} / \text{No}) = 0.0206$

$$V_{NB}(\text{yes}) = \frac{V_{NB}(\text{Yes})}{V_{NB}(\text{yes}) + V_{NB}(\text{No})} = 0.205$$

$$V_{NB}(\text{No}) = \frac{V_{NB}(\text{No})}{V_{NB}(\text{yes}) + V_{NB}(\text{No})} = 0.795$$

$V_{NB}(\text{No}) > V_{NB}(\text{Yes}) \Rightarrow \text{play} = \text{No}$